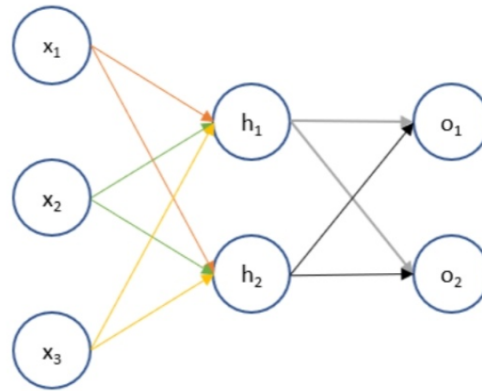


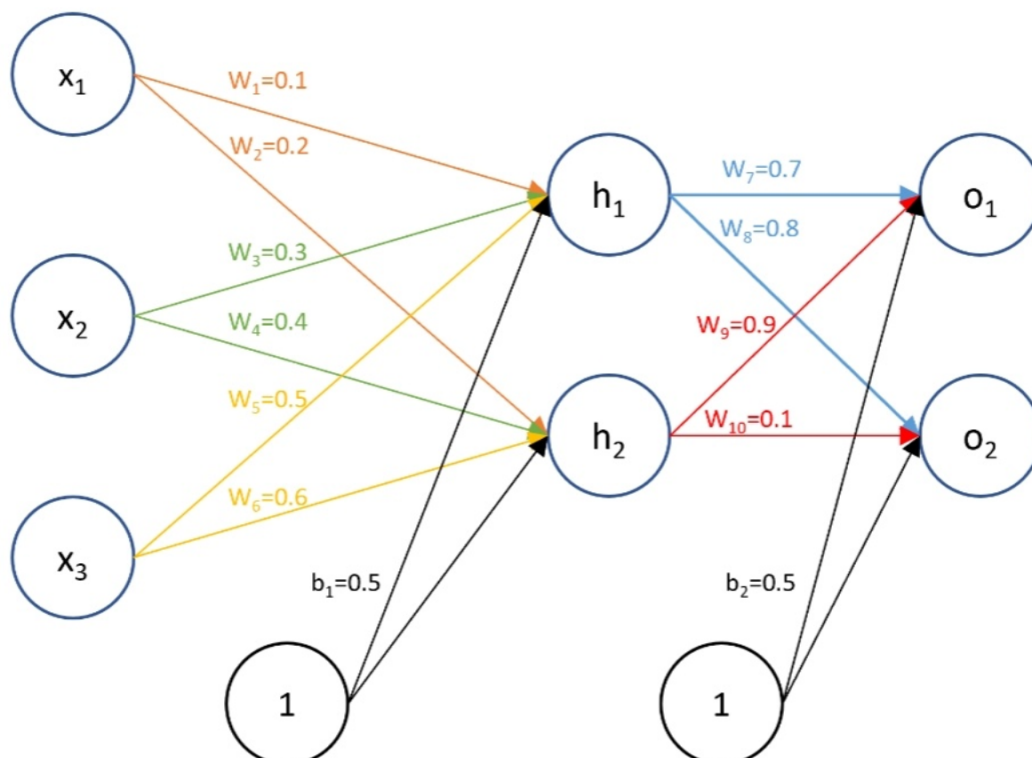
Backpropagation by Hand



Assume we have the above network to be trained. The following are the (very) high level steps to train this network.

- (1) Initialize weights for the parameters we want to train
- (2) Forward propagate through the network to get the output values
- (3) Define the error or cost function and its first derivatives
- (4) Backpropagate through the network to determine the error derivatives
- (5) Update the parameter estimates using the error derivative and the current value

1. Initialize



The input and target values for this problem are $x_1=1, x_2=4, x_3=5$ and $t_1=0.1, t_2=0.05$. Initialize weights as shown in the diagram. Generally, you will assign them randomly but for calculation purposes, I've chosen these numbers.

Inputs:

$$\cdot x_1 = 1, x_2 = 4, x_3 = 5$$

Targets:

$$\cdot t_1 = 0.1, t_2 = 0.05$$

Weights:

$$\cdot \text{Input} \rightarrow \text{Hidden: } w_1 = 0.1, w_2 = 0.2, w_3 = 0.3, w_4 = 0.4, \\ w_5 = 0.5, w_6 = 0.6$$

$$\cdot \text{Output} \rightarrow \text{Hidden: } w_7 = 0.7, w_8 = 0.8, w_9 = 0.9, w_{10} = 0.1$$

$$\text{Biases: } b_1 = 0.5, b_2 = 0.5$$

$$\text{Activation: } \sigma(z) = \frac{1}{1 + e^{-z}}$$

* Forward Propagate:

(A) Input $\rightarrow$ Hidden

$$\begin{aligned} zh_1 &= 0.1 \times 1 + 0.3 \times 4 + 0.5 \times 5 + 0.5 \\ &= 0.1 + 1.2 + 2.5 + 0.5 \\ &= 4.3 \end{aligned}$$

$$\begin{aligned} zh_2 &= 0.2 \times 1 + 0.4 \times 4 + 0.6 \times 5 + 0.5 \\ &= 0.2 + 1.6 + 3.0 + 0.5 \\ &= 5.3 \end{aligned}$$

$$h_1 = \sigma(zh_1) = \sigma(4.3) = \frac{1}{1 + e^{-(4.3)}} = \frac{1}{1 + 2.71}^{(4.3)}$$

$$= \frac{1}{1 + 0.014} = 0.9866$$

$$h_2 = \sigma(z_{h_2}) = \sigma(5.3) = \frac{1}{1 + 0.01} = 0.9950$$

(B) Hidden $\rightarrow$ Output

$$\begin{aligned} z_{o_1} &= w_7 h_1 + w_8 h_2 + b_2 \\ &= 0.7 \times 0.9866 + 0.9 \times 0.9950 + 0.5 \\ &= 2.0862 \end{aligned}$$

$$\begin{aligned} z_{o_2} &= w_9 h_1 + w_{10} h_2 + b_2 \\ &= 0.8 \times 0.9866 + 0.1 \times 0.9950 + 0.5 \\ &= 1.3888 \end{aligned}$$

$$o_1 = \sigma(z_{o_1}) = \sigma(2.0862) = 0.8896$$

$$o_2 = \sigma(z_{o_2}) = \sigma(1.3888) = 0.8004$$

★ Sum of Squares Error :

Define the sum of squares error using the target values and the results from the last layer from forward propagation.

$$E = \frac{1}{2}[(o_1 - t_1)^2 + (o_2 - t_2)^2] = \frac{1}{2}[(0.8896 - 0.1)^2 + (0.8004 - 0.05)^2] = 0.5932$$

$$\frac{dE}{do_1} = o_1 - t_1 = 0.8896 - 0.1 = 0.7896$$

$$\frac{dE}{do_2} = o_2 - t_2 = 0.8004 - 0.05 = 0.7504$$

* Back propagate

(A) Output layer "deltas"

$$\frac{\partial E}{\partial o_1} = o_1 - t_1, \quad \frac{\partial o_1}{\partial z_{o_1}} = o_1 (1 - o_1)$$

$$\delta_{o_1} = \frac{\partial E}{\partial z_{o_1}} = (o_1 - t_1) o_1 (1 - o_1) = 0.0776$$

Similarly:

$$\delta_{o_2} = \frac{\partial E}{\partial z_{o_2}} = (o_2 - t_2) o_2 (1 - o_2) = 0.1199$$

(B) Gradients for w_7, w_8, w_9, w_{10} & b_2

$$\cdot z_{o_1} = w_7 h_1 + w_8 h_2 + b_2$$

$$\cdot z_{o_2} = w_9 h_1 + w_{10} h_2 + b_2$$

$$\frac{\partial E}{\partial w_7} = \delta_{o_1} h_1 = 0.0765$$

$$\frac{\partial E}{\partial w_9} = \delta_{o_1} h_2 = 0.0772$$

$$\frac{\partial E}{\partial b_2} = \delta_{o_1} + \delta_{o_2} = 0.1975$$

(C) Backprop to hidden layer

$$\frac{\partial E}{\partial h_1} = \delta_{o_1} w_7 + \delta_{o_2} w_9 = 0.1502$$

$$\frac{\partial E}{\partial h_2} = \delta_{o_1} w_8 + \delta_{o_2} w_{10} = 0.0818$$

$$\delta_{h_1} = \frac{\partial E}{\partial z_{h_1}} = \frac{\partial E}{\partial h_1} h_1 (1-h_1) = 0.0020$$

$$\delta_{h_2} = \frac{\partial E}{\partial z_{h_2}} = \frac{\partial E}{\partial h_2} h_2 (1-h_2) = 0.0004$$

(1) Gradients for input $\rightarrow$ hidden weights and b ,

$$\cdot z_{h_1} = w_1 x_1 + w_3 x_2 + w_5 x_3 + b_1$$

$$\cdot z_{h_2} = w_2 x_1 + w_4 x_2 + w_6 x_3 + b_1$$

$$\frac{\partial E}{\partial w_1} = \delta_{h_1} x_1 = 0.0020, \quad \frac{\partial E}{\partial w_3} = \delta_{h_1} x_2 = 0.0079$$

$$\frac{\partial E}{\partial w_5} = \delta_{h_1} x_3 = 0.0049$$

$$\frac{\partial E}{\partial w_2} = \delta_{h_2} x_1 = 0.004, \quad \frac{\partial E}{\partial w_4} = \delta_{h_2} x_2 = 0.0016$$

$$\frac{\partial E}{\partial w_6} = \delta_{h_2} x_3 = 0.0020$$

$$\frac{\partial E}{\partial b_1} = \delta_{h_1} + \delta_{h_2} = 0.0024$$

* Update parameters (learning rate $\alpha = 0.01$)

Rule:

$$w \leftarrow w - \alpha \frac{\partial E}{\partial w},$$

$$b \leftarrow b - \alpha \frac{\partial E}{\partial b}$$

Updated

$$\cdot w_1 = 0.099980$$

$$\cdot w_2 = 0.199996$$

$$\cdot w_3 = 0.299921$$

$$\cdot w_4 = 0.399984$$

$$\cdot w_5 = 0.499901$$

$$\cdot w_6 = 0.599980$$

$$\cdot w_7 = 0.699235$$

$$\cdot w_8 = 0.798817$$

$$\cdot w_9 = 0.899228$$

$$\cdot w_{10} = 0.098807$$

$$\cdot b_1 = 0.499976$$

$$\cdot b_2 = 0.498025$$